

Ian Noronha

West Lafayette, IN | +1 (765)-772-8344 | inoronha@purdue.edu | [Linkedin](#) | [Google Scholar](#) | [Website](#)

EDUCATION

Purdue University

Graduate Researcher, Vision for Robotics — GPA: 3.89/4.0

West Lafayette - IN, USA

Aug 2024 – Ongoing

Indian Institute of Technology, Roorkee (IIT Roorkee)

Bachelor of Technology in Electronics and Communication Engineering

Roorkee, India

2019

EXPERIENCE

ARIES Lab, Purdue University

Graduate Research Assistant

West Lafayette - IN, USA

Aug 2024 – Present

- Mentored 12 undergraduate and graduate researchers at the ARIES Lab, Purdue University, on projects spanning neuromorphic sensing and intelligent vision systems.
- Leading research on neuromorphic perception, uncertainty-aware sensor fusion for autonomous robot navigation, self-supervised reinforcement learning, 3D scene-aware Vision-Language Model agents that bridge spatial reasoning and natural language understanding for real-world autonomous systems.

STMicroelectronics

Design Engineer, Advanced Systems Research (ASR), R&D – Edge AI Applications

Greater Noida, India

Apr 2021 – June 2024

- Led research in Robotics & Vision, received CEO's commendation for contributions to embodied AI research.
- Mapped ViT layers onto In-Memory-Compute tiles, achieving a 40% improvement in on-device accuracy.
- Trained and deployed depth estimation models for indoor SLAM on STM32N6 and STM32-MP1 hardware.
- Mentored 2 students during their industry co-op. Developed custom Linux kernels running ROS and ORB-SLAM3 for Cortex A7 processor boards; developed neural-network-to-C compilers with partners at IBM and ETH Zurich.

Robotics and Control Lab, IIT Roorkee

Research Fellow

Roorkee, India

Mar 2020 – Apr 2021

- Led the end-to-end development of a voice-controlled, non-contact patient registration system.
- Successfully transitioned the registration system from prototype to product; deployed at a Hospital.
- Contributed to the founding of Robolife Ltd., a faculty enterprise by Prof. P.M. Pathak valued at \$800k USD

Deloitte USI

Business Technology Analyst

Hyderabad, India

June 2019 – Mar 2020

- Engineered automation tools for data archival for JP Morgan Chase using Python, X-Query, and SQL.
- Developed a custom drone-based mapping application for a strategic firm initiative.

PATENTS

- [1] **mmWave Radar system with AI for Sub-Wavelength Detection of Hidden Small Organisms**
US Provisional Patent | Inventors: U. Kaur, **I. Noronha**, S. Bharti, M. Murgia, C.Hill
- [2] **Radar-based system and method for detecting structural anomalies in granular materials**
US Provisional Patent | Inventors: U. Kaur, **I. Noronha**, S. Bharti, R.P.K. Ambrose
- [3] **Untethered Soft Snake Robot.** IN 492402 — Granted Jan 2024
Inventors: G. Bhandari, Y. Pirjade, P.M. Pathak, **I.W. Noronha**, J.M. Yang
- [4] **Non-contact Patient Registration and Monitoring System.** IN 479029 — Granted Dec 2023
Inventors: P.M. Pathak, **I.W. Noronha**

PUBLICATIONS

Published/Accepted Peer-Reviewed Conferences

- [1] MooCap: A Multi-View Benchmark for Cow-Object-Human Interaction and Behaviour Dynamics
I. Noronha, H.W. Neave, U. Kaur CVPR 2026
- [2] Efficient Event Camera Volume System
J.C. Soto*, **I. Noronha***, S. Bharti, U. Kaur ICRA 2026

- [3] Graph Coloring for Multi-Task Learning
S. Patapati, I. Noronha ECCV 2026
- [4] MBE-ARI: A Multimodal Dataset Mapping Bi-Directional Engagement in Animal-Robot Interaction
*I. Noronha**, *A.P. Jawaji**, *J.C. Soto**, *J. An*, *Y. Gu*, *U. Kaur* ICRA 2025
- [5] Deeply Quantized Neural Networks: Enabling In-Sensor Computing
M. Chandna, *S. Suneja*, *L. Luise*, *F.V. Caruso*, *I.W. Noronha*, *H.D. Dabral*, *S.P. Singh* ISTEMS 2024
- [6] nn-morph: Simplifying Deep Neural Network Manipulation, Transformation, and Analysis
I.W. Noronha, *S. Suneja*, *S.P. Singh* ICCCNT 2023

Workshop/Poster presentations

- [1] Quad Forecaster: Diffusion-Based Quadruped Pose Prediction for Animal Communication Analysis
*I. Noronha**, *A. Chowdhury**, *S. Bharti*, *U. Kaur* NeurIPS Workshop 2025
- [2] Spatiotemporal Representations for Event Cameras: Towards Seamless Integration in Robotic Perception
I. Noronha, *J.C. Soto*, *U. Kaur* ICRA Workshop 2025
- [3] Feel the Ground: Radar based Terrain Classification towards Gait Adaptation in Legged Robots
S. Bharti, *I. Noronha*, *Gabriel Torres*, *U. Kaur* IROS LBR 2026
- [4] FieldSense: Real-Time Plant Disease Detection for Precision Agriculture (Oral)
*S. Bharti**, *I. Noronha**, *J.C. Soto**, *A. Raturi*, *D.R. Buckmaster*, *U. Kaur* ASABE 2026
- [5] MM-Wave Radar Identification of Cavities in Grain Storage Systems to Prevent Entrapment (Oral)
S. Bharti, *I. Noronha*, *A. Agarwal*, *S.N. Doctor*, *R.P.K. Ambrose*, *U. Kaur* ASABE 2026

Submitted Manuscripts Under Review

- [1] E-motion: High Fidelity Event Generation from Frame-Based Input for Neuromorphic Vision
I. Noronha, *J.C. Soto*, *S. Bharti*, *S. Singh*, *P. Chitivelu*, *J. Wachs* *U. Kaur* WACV 2026
- [2] Beyond the Turing Test: A Game Theoretic Approach to LLM-Assisted Submissions
S. Bharti, *I. Noronha*, NeurIPS 2026
- [3] Elevation from Range: Cold-start mmWave Radar-Camera Calibration
*I. Noronha**, *S. Bharti**, *Gabriel Torres*, *U. Kaur* RA-L
- [4] SemanticSkeleton: Canonizing Quadruped Skeletons for Cross-Skeletal Pose Forecasting
A. Chowdhury, *I. Noronha[†]*, *U. Kaur* WACV 2026

* Denotes equal contribution [†] Denotes Project lead

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Embedded C, CUDA, Java
- **ML Frameworks:** PyTorch, TensorFlow, JAX, TensorRT, ONNX Runtime, STM32Cube.AI, Weights & Biases
- **Robotics & Perception:** ROS/ROS2, OpenCV, SLAM, Sensor Fusion, 3D Point Cloud Processing, Stereo Vision, Depth Estimation, Event Cameras/Neuromorphic Vision, Spiking Neural Networks
- **Planning, Control & Simulation:** Model Predictive Control (MPC), Reinforcement Learning (RL), Gazebo, NVIDIA Isaac Sim, MATLAB/Simulink, VLM/VLA planning
- **Hardware & Edge Deployment:** NVIDIA Jetson, STM32, Raspberry Pi, TI mWave Radars, Synsense Speck/Xylo
- **Dev & MLOps:** Git, Docker, CI/CD, Linux Kernel Development

HONORS & AWARDS

- **1st Place**, SyDAG Hackathon (International) Oct 2025
- **3rd Place**, IEEE COINS Tiny ML Contest (International) Jul 2023
- **Special Recognition Award**, STMICROELECTRONICS Tech Week Hackathon (International) Mar 2023
- **3rd Place**, WiFi Thinkfest (National) Apr 2018
- **1st Place**, Inter IIT Technical Meet (National), Project: Safety Device for Fishing Vessels Jan 2018
- **2nd Place**, Pitney Bowes SMB Hackathon (International) Jan 2018
- **1st Place**, Srishti IIT Roorkee Technical Fest, Project: Interactive VR in Gaming Jul 2017

SERVICE & LEADERSHIP

- **Reviewer:** IROS 2025, ICRA 2026, NeurIPS 2026
- **Co-Founder & Secretary, Culinary Club, IIT Roorkee:** Organized events with 100+ participants; introduced the first student-run cafeteria in IIT. Spearheaded National events at Inter-IIT Cultural meets with participants from all over India
- **Addl. Secretary, AI & Electronics Society, IIT Roorkee:** Led a 300-member society to organize lectures and workshops.